Recently, we studied a new type of test that tests the hypothesis that ALL the coefficients in the multiple linear regression model are equal to zero.

new test for $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$

One idea we didn't address when studying that test is why we would use it at all, when we already knew how to perform individual tests of whether each coefficient equals 0.

how is that different from doing multiple tests?

$H_0: \beta_1 = 0$, $H_0: \beta_2 = 0$ ... , $H_0: \beta_p = 0$

Much like fitting one multiple linear regression model is different from fitting several different simple linear regression models, performing a bunch of individual tests is not the same as performing one overall test.

(it IS different!)

We'll briefly explain WHY it isn't the same, but you are only required to know that it isn't the same, not the explanation.

hypothesis test w/5% rejection threshold:
when H_0 is TRUE, we (mistakenly) reject 5% of the time

When we perform hypothesis tests, there is always a chance that we reject H_0, when H_0 was actually true. Specifically, that chance is equal to the rejection threshold (α).

If we perform an individual hypothesis test for each of the

coefficients $\beta_1, \beta_2, \ldots, \beta_p$, then each one of those tests can mistakenly reject a TRUE null hypothesis 5% of the time (or whatever other % we used as the rejection threshold).

if truly $\beta_1 = \beta_2 = \ldots = \beta_p = 0$,

then each test $H_0: \beta_k = 0$ mistakenly rejects 5% of the time

But the probability that at LEAST one of the set of indiv tests falsely rejects is greater than 5%.

but probability that at LEAST one test mistakenly rejects is $> 5\%$

If 13% of people are left handed, and we sample 10 people, then for each individual person we sample, there is a 13% chance we'll get a lefthanded person. But the probability that at least one of the 10 people is left-handed is NOT 13%, it's bigger.

P(left handed) = .13

P(≥1 left handed person in sample of 10) = $1 - P(0 \text{ left-handed})$ (≈75%)

$= 1 - (0.87)^{10}$

So if we want to check whether all of the slopes in the multiple linear regression model equal 0, and we do that by performing many separate tests of individual coefficients with a 5% rejection threshold, then the probability that we mistakenly conclude there IS some linear relationship, when truly there is none, can be a lot bigger than 5%.

Suppose there are 10 predictors $X_1, X_2, \ldots, X_{10}$
and truly $\beta_1 = \beta_2 = \ldots = \beta_{10} = 0$
(no linear assoc btwn predictors & response)

we test: $H_0: \beta_1 = 0$, ..., $H_0: \beta_{10} = 0$
5% chance of mistake for EACH test

If we reject the null hypothesis in any of these tests, then we would incorrectly conclude that there IS a linear assoc btwn the response(y), and at least one of the predictor variables.
chance of ≥ 1 mistake overall $> 5\%$

When we use a single, combined test of the hypothesis
$H_0: \beta_1 = \ldots = \beta_{10} = 0$ w/ rejection threshold 5%
to check whether there is any linear association, this test is designed to reject a true null hypothesis 5% of the time
(or whatever % we use as the rejection threshold)
if H_0 true, F test mistakenly rejects 5% of the time

That is the reason we use a new type of test for this hypothesis, rather than looking at the individual p-values associated with each predictor variable in the model summary output.

Now, we can perform a single test to check whether a group of predictors, or ALL of the predictors, have coefficients of zero.
These tests are based on the idea that if certain predictors don't provide any additional info about the response, then removing them from the model shouldn't make a big difference in terms of how well the model predicts the response(y).

F tests

eg $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$

$H_0: \beta_2 = \beta_3 = 0$ means x_2 & x_3 provide no EXTRA info about y (when x_1 is included)

↓

if true, removing x_2 & x_3 from model shouldn't worsen predictions very much

For the restaurant price data, we considered modeling price as a function of food quality, service quality and decor. Then we asked whether the "service" and "decor" variables truly helped the linear model do a better job of predicting price.

Compare 2 models:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon$ "full"

$y = \beta_0 + \beta_1 x_1 + \varepsilon$ "reduced"

To answer that question, we compared 2 models: a "full" model that used all the predictors, and a "reduced" model that omitted the predictors with coefficients of zero in the null hypothesis.

The quantity we compare is the sum of squared residuals for each of the models

Compare sum of squared residuals: RSS_{red} & RSS_{full}

If the omitted predictors (service and decor) DO provide addtl information about price, (in the linear model w/ food quality), then omitting them would result in worse predictions for y (price), and a larger sum of squared residuals in the reduced model.

- if H_0 false → reduced model should be WORSE
  $RSS_{red} > RSS_{full}$
  if H_0 true → $RSS_{red} \approx RSS_{full}$

To quantify this, we calculated a test statistic:

$$f_{obs} = \frac{(RSS_{red} - RSS_{full})/(df_{red} - df_{full})}{RSS_{full}/df_{full}}$$

where df_{full} is the degrees of freedom for the full model, $n-p-1$
and df_{red} is the df for the reduced model, $n - \{\#\text{ predictors in reduced model}\} - 1$

A large value of the test statistic indicates that omitting those predictors from the model makes it much worse, which would lead us to reject the null hypothesis that those coefficients (β_2, β_3) equal zero.

large value of f_{obs} suggests removing x_2 & x_3 makes model much worse

↳ incompatible w/ $H_0: \beta_2 = \beta_3 = 0$

This test is not something we'll perform manually - instead, use built-in functions (eg using "anova( )" function in R).

★ GO TO NOTEBOOK/SLIDES ★

Something we did not specifically discuss last time is that the sum of the squared residuals for the reduced model will NEVER be smaller than the sum of squared residuals for the full model.

note: $RSS_{red} \geq RSS_{full}$, always

Thinking of this in the opposite direction, when we ADD a predictor variable to a linear regression model, the sum

of squared residuals can never go UP.

add new predictor to model → sum of squared residuals goes down (or stays same).

Including a new predictor will never give us a model that fits the observed data worse. Why not?

Suppose we have ~~a~~ a data set with ~~three~~ two predictors;

suppose we have y, x_1, x_2, x_3

If we fit a model $\hat{y} = \hat{b}_0 + \hat{b}_1 x_1 + \hat{b}_2 x_2$, the coefficients are chosen to give the smallest possible sum of squared residuals.

$\hat{b}_0, \hat{b}_1, \hat{b}_2$ minimize $\sum_{i=1}^{n} (y_i - (\hat{b}_0 + \hat{b}_1 x_{i1} + \hat{b}_2 x_{i2}))^2 = RSS = SSR_A$.

If we fit a model using both predictors, it looks like:

$\hat{y} = \hat{b}_0 + \hat{b}_1 x_1 + \hat{b}_2 x_2$, and in this model the coefficients are ALSO chosen to minimize the sum of squared residuals.

$\hat{b}_0', \hat{b}_1', \hat{b}_2'$ minimize $\sum_{i=1}^{n} (y_i - (\hat{b}_0' + \hat{b}_1' x_{i1} + \hat{b}_2' x_{i2}))^2 = RSS = SSR_B$

We can always get this second sum, RSS = SSR_B, to be at LEAST as small as the first one, because we could choose the SAME intercept and slope b_1, and just set $\hat{b}_2 = 0$.

When we choose the BEST coefficients for the second model, the coefficients from the first model are an option, so we'll never pick anything WORSE than that.

if we choose $\hat{b}_0' = \hat{b}_0$, $\hat{b}_1' = \hat{b}_1$, $\hat{b}_2' = 0$, $RSS_{SSR_B} = RSS_{SSR_A}$

. of squared residuals can never go up.

add new predictor to model → sum of squared residuals goes down (or stays same)

Including a new predictor will never give us a model that fits the observed data worse. Why not?

Suppose we have a data set with two predictors: y_i, x_{i1}, x_{i2}. If we fit a model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$, the coefficients are chosen to give the smallest possible sum of squared residuals.

$\hat{\beta}_0, \hat{\beta}_1$ minimize $\sum_{i=1}^{n} (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_{i1}))^2 = RSS_A$

If we fit a model using both predictors, it looks like: $\hat{y} = \hat{\beta}'_0 + \hat{\beta}'_1 x_1 + \hat{\beta}'_2 x_2$, and in this model the coefficients are ALSO chosen to minimize the sum of squared residuals.

$\hat{\beta}'_0, \hat{\beta}'_1, \hat{\beta}'_2$ minimize $\sum_{i=1}^{n} (y_i - (\hat{\beta}'_0 + \hat{\beta}'_1 x_{i1} + \hat{\beta}'_2 x_{i2}))^2 = RSS_B$

We can always get this second sum, RSS_B, to be at least as small as the first one, because we could choose the SAME intercept and slope β_1, and just set $\hat{\beta}'_2 = 0$.

When we choose the BEST coefficients for the second model, the coefficients from the first model are among the options, so we'll never pick anything WORSE than that.

If we choose $\hat{\beta}'_0 = \hat{\beta}_0$, $\hat{\beta}'_1 = \hat{\beta}_1$, $\hat{\beta}'_2 = 0$

then $RSS_B = RSS_A$

- if DIFFERENT values of $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)$ give smaller RSS_B ✓
  choose those instead.

  if NOT, can still match RSS_A

So when we compare a "full" model and a "reduced" model, the sum of squared residuals in the full model will always be smaller, or the same. Never larger.

$$f_{obs} = \frac{(SS_{red} - SS_{full})/(df_{red} - df_{full})}{SS_{full}/df_{full}} \geq 0$$

This fact has another notable consequence.

In multiple linear regression models, we can still calculate R^2:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

and it still represents the % of variation in y values explained by model.

When a new predictor is added to the model, the sum of squared residuals goes down (or stays the same).

add predictor → $\sum(y_i - \hat{y}_i)^2$ goes down (or stays same)

The amount of unexplained variation in y goes down, in other words. Correspondingly, the R^2 value goes up (or stays same).

→ R^2 goes up (or stays same)

R^2 is meant to be a measure that tells us how well the model fits. So it seems, based on R^2, that including more predictors always makes the model fit better.

add predictor → model fits better ???

But R^2 ALSO goes up if we add a predictor whose coefficient in the population model equals zero.

what if true $\beta_k = 0$? adding x_k really "better"?

For any particular finite set of sample data, the sample correlation between x_k and y will not be exactly zero.

if $\beta_k = 0$, SAMPLE correl btwn x_k and y won't be exactly 0 (due to random chance)

(just like for any random variable, the sample mean won't exactly equal the population mean)

so in the sample data, there will be some amount of association btwn x_k and y, purely by random chance, and R^2 will improve when we add x_k to the model.

so adding x_k to model will increase R^2

But that is misleading! In the overall population, x_k doesn't actually provide any extra info about y (in this particular linear model).

misleading! including x_k is NOT truly better, $\beta_k = 0$

Remember that our ultimate goal is to estimate the true population relationship, not just the relationship in this one particular set of sample data.

To correct for this problem, i.e. that adding predictors to the model always makes the R^2 value go up, there is an "adjusted" version of R^2.

more useful: "adjusted" R^2
- takes into account # predictors
- CAN go down when irrelevant predictor is added
- allows fair comparison of "full" and "reduced" models

The "adjusted" version takes into account how many predictors are actually used in the model, adjusting for the fact that R^2 would otherwise be larger w/more predictors, just due to chance.

This lets us meaningfully compare "full" and "reduced" models.
The adjusted R^2 value CAN ~~go~~ decrease if an irrelevant predictor is added to the model.

If we add a predictor and the adjusted R^2 value decreased, that can be an indication that the new predictor is not useful.

As the analyst, you get to choose which predictors to include in the model. You don't necessarily need to use every predictor variable you have access to.

which predictors to include in model? up to you

The adjusted R^2 value is one tool you could consider when deciding whether or not to include a variable.

It may be tempting to use p values to decide which predictors to use – fit a model w/all of them, then eliminate the ones whose slope coefficients were not significantly different from zero. This is generally NOT considered to be best practice.

don't: only include predictors w/ slope signif. different from 0
(i.e. for which ~~$H_0: b_k = 0$ has small~~ we reject $H_0: \beta_k = 0$)

Failing to reject the null hypothesis that $\beta_k = 0$ doesn't mean we accept that b_k definitely IS zero.

(fail to reject $H_0: \beta_k = 0$ ≠ accept that $\beta_k = 0$ is true)

If we wanted to use adj R^2 to decide which predictors to include, how might we do that?

variable selection

① fit one model for each combination of predictors

If we only have a few possible predictor variables, we could fit a different regression model for every subset of those predictors.

eg for restaurant prices:

food	food + service	all 3
service	food + decor	
decor	service + decor	

For the restaurant price data, we could fit a separate model for every possible combination of (food, service, decor). That would be 7 different models, which we can fit and compare. If we have a large number of predictors, this is impractical - we'd have to fit a very large # of models.

impractical w/large # of predictors → too many possible combinations to check them all

what could we do instead?

② plots: scatterplots of response & each predictor

It can be helpful to look at scatterplots of the response with each predictor individually.

If the scatter plot of x_k and y does not show an association, then x_k will probably not be helpful for predicting y in a multiple linear regression model either.

no obvious assoc btwn x_k and y in plot → x_k probably not helpful for predicting y

You could decide to exclude some of the predictor variables based on that informal check.

It's also fairly common to take a ③ "stepwise" approach to selecting variables. The process goes like this:

start by fitting the full model, using all the predictors (# = p)

* fit full model
  * for each predictor, check whether removing it increases adj R^2
  * remove predictor that gives largest increase in adj R^2 (if any) This is the new model
* for new model, again try removing each predictor.
  * if removing a predictor increases adj R^2, choose the one that gives largest increase

iterate until removing predictors only decreases adj R^2

For each predictor, we remove that predictor and fit a new model. Among the new models with p-1 predictors, choose the one with the biggest increase in adj R^2 compared to the full model. (If none are an improvement, stop and use full model).

Then repeat the process: if removing a predictor from the new model increases adj R^2, remove the predictor that gives the biggest increase.

Keep going until removing predictors only decreases adj R^2.

In each "step", we permanently remove <u>one</u> predictor from the model.

We will not practice implementing this - the R and python tools for stepwise variable selection aren't the most user-friendly (in my opinion). But this could potentially be useful in future data analysis projects, so it's good to know that it exists.

Another note - when choosing predictor variables to include, there are other selection criteria aside from adj R^2.

variable selection criteria: adj R^2 is not the only one ("AIC", "BIC")